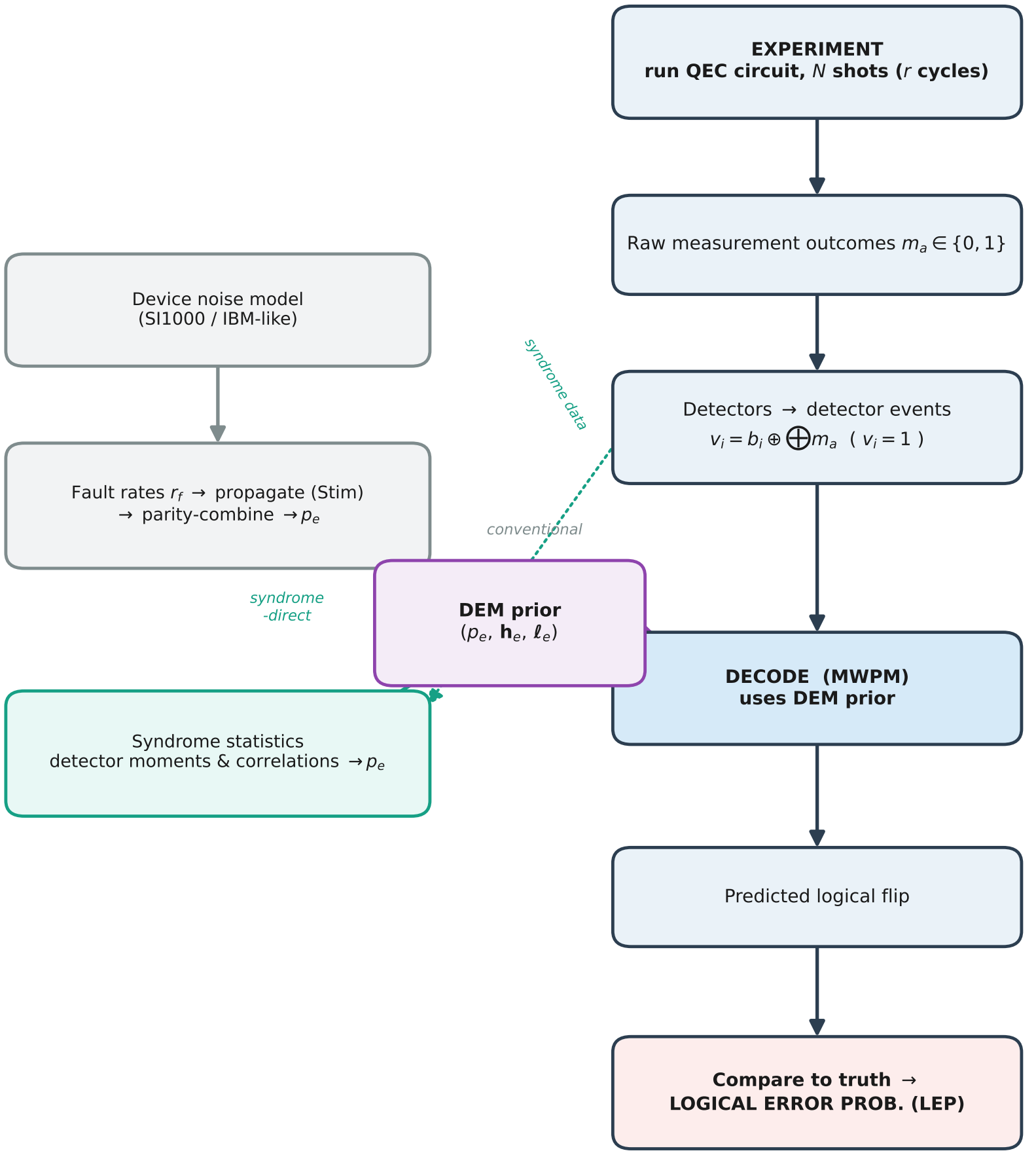
From experiment to logical error: where the three DEM papers act



What each paper does

1 Blume-Kohout & Young (2504.14643)

THEORY. DEM is the uniquely learnable noise object; Walsh-Hadamard formalism; meaning of p_{ij}. Underpins syndrome est.

2 Remm et al. (PRX Res. 2026)

EXPERIMENT, $d = 3$. Characterize error channels (Y, hook, leakage ≤ 8 cycles) from syndrome corr.; set MWPM weights in-situ.

3 Takou...Brown (2606.11496)

EXPERIMENT, Willow + IBM. Compare prior sources (device vs RL vs syndrome-est.); measure LEP reduction (~5-10%).

Decoder prior = the DEM. Conventional: device model → r_f → p_e (needs benchmarking). Syndrome-direct: estimate p_e from detector statistics.